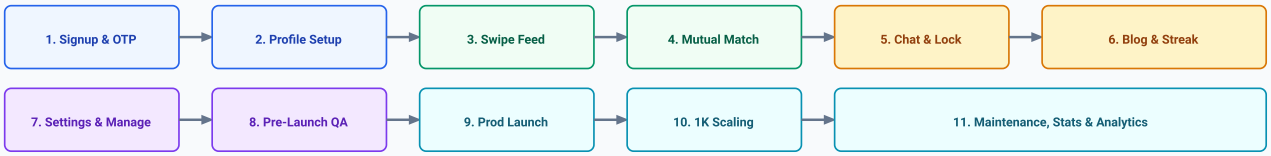
MASTER EXECUTION MAP

Skill Exchange Platform Architecture

End-to-End Task Breakdown, Role Matrix & Operational Flowchart

Project Lead: Frontend + Backend
Target: 1,000 Active Users

1. COMPLETE END-TO-END LIFECYCLE FLOWCHART



STAGE 1: REGISTRATION, AUTHENTICATION & PHONE OTP VERIFICATION

Frontend Lead (UI / Client)

- Build multi-step Signup/Login screen with client-side validation.
- Design 6-digit auto-advancing OTP pin input with SMS resend timer.
- Implement JWT secure storage handling in browser storage/cookies.

Backend Lead (Server / DB)

- Configure authentication API endpoints for email and password.
- Integrate Twilio SMS API to route real-time 6-digit verification codes.
- Construct user authorization middleware and verify OTP hashes.

STAGE 2: PROFILE ONBOARDING & SKILL TAXONOMY SETUP

Frontend Lead (UI / Client)

- Create interactive Skill Selector tag input for offered vs. desired skills.
- Build profile photo upload preview and bio editor forms.
- Implement profile preview card prior to deck inclusion.

Backend Lead (Server / DB)

- Design PostgreSQL schema for Users, Skills, and UserSkills join tables.
- Implement file storage bucket integration for profile avatars.
- Build profile update API endpoints with schema validation.

STAGE 3: SWIPE FEED & QUOTA ENGINE (10 VS 50 SWIPES)

Frontend Lead (UI / Client)

- Implement drag-and-swipe deck mechanics using Framer Motion gestures.
- Display real-time quota counter widget (10 Free / 50 Paid swipes).
- Build "Quota Reached" banner with modal prompting streak extensions.

Backend Lead (Server / DB)

- Develop matching recommendation feed algorithm (Reciprocal Skill Filter).
- Build swipe submission API logging Right (Pass) and Left (Interest) actions.
- Implement automated daily cron job resetting swipe quotas at 00:00 UTC.

STAGE 4 & 5: MUTUAL MATCH ENGINE & ACTIVE CHAT LOCK

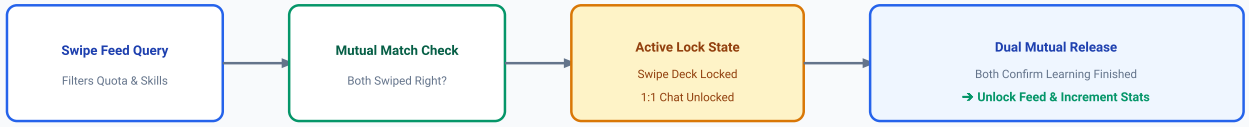
Frontend Lead (UI / Client)

- Design "It's a Barter Match!" full-screen overlay popup animation.
- Build real-time 1:1 messaging window with chat bubble UI components.
- Implement "Skill Barter Completed" mutual agreement modal button.

Backend Lead (Server / DB)

- Build match detection trigger executing on mutual right-swipes.
- Deploy WebSocket (Socket.io) server for instant real-time message delivery.
- Implement Exchange Lock engine freezing swipe API access until session completion.

DISCOVERY, MATCH & EXCHANGE LOCK STATE ARCHITECTURE



STAGE 6: BLOG SECTION & DAILY STREAK GAMIFICATION

Frontend Lead (UI / Client)

- Build Markdown rich text editor for writing micro learning logs.
- Create community Blog Feed view with skill tag filtering.
- Design active Streak Counter widget displaying streak bonus swipes.

Backend Lead (Server / DB)

- Create CRUD endpoints for blog posts, comments, and upvotes.
- Implement scheduled daily worker analyzing post frequency per user.
- Build automated bonus swipe allocator adding extra feed credits for streaks.

STAGE 7: SETTINGS DASHBOARD & ACCOUNT GOVERNANCE

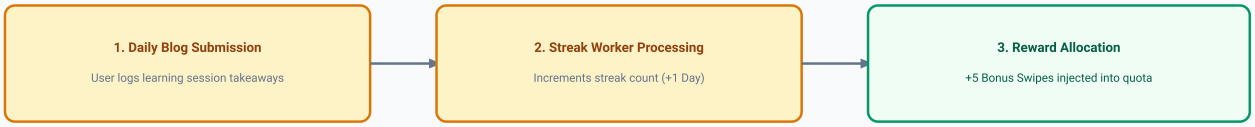
Frontend Lead (UI / Client)

- Build profile settings forms for altering skills, bio, and phone number.
- Design notification preference toggle switches.
- Build Account Deletion confirmation modal with double-confirmation prompt.

Backend Lead (Server / DB)

- Build profile update API validating schema and phone re-verification.
- Create permanent cascade account deletion endpoint purging user rows.
- Implement token revocation and S3 asset cleanup upon deletion.

GAMIFICATION & ACCOUNT LIFECYCLE ARCHITECTURE



STAGE 8: PRE-LAUNCH QUALITY ASSURANCE & TESTING SUITE

Frontend Lead (UI / Client)

- Perform cross-browser and cross-device responsive layout testing.
- Conduct end-to-end user testing for swipe gesture fluid performance.
- Audit web accessibility and light/dark theme visual fidelity.

Backend Lead (Server / DB)

- Write integration unit test suite for swipe matching logic.
- Execute API stress testing under high synthetic concurrent requests.
- Perform security penetration audit checking WebSocket auth leaks.

STAGE 9: PRODUCTION DEPLOYMENT & INFRASTRUCTURE LAUNCH

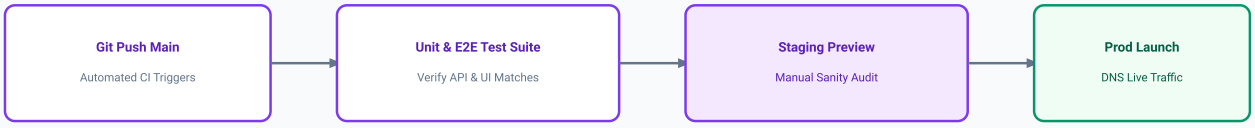
Frontend Lead (UI / Client)

- Deploy production React build to Vercel global CDN edge network.
- Configure custom SSL domain route with optimized asset compression.
- Set up Sentry error monitoring on client application runtime.

Backend Lead (Server / DB)

- Provision production Node server instance on Render with auto-scaling.
- Configure managed PostgreSQL database instance on Supabase.
- Setup environment secret keys, SSL certificates, and CORS rules.

PRE-LAUNCH CI/CD & STAGING VALIDATION PIPELINE



STAGE 10: INFRASTRUCTURE SCALING FOR FIRST 1,000 USERS

Frontend Lead (UI / Client)

- Implement optimistic UI updates for instant swipe responsiveness.
- Configure lazy-loading for profile media and blog assets.
- Minimize JavaScript bundle footprint with code splitting.

Backend Lead (Server / DB)

- Create DB indexes on `skills_offered` and `skills_desired` columns.
- Configure Redis caching layer for active user swipe decks.
- Setup database connection pooling to handle concurrent WebSocket connections.

STAGE 11: COMMUNITY MODERATION & USER SUPPORT SYSTEM

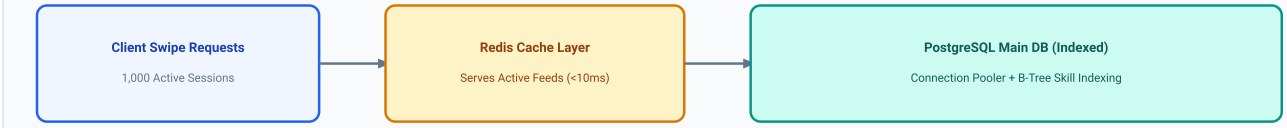
Frontend Lead (UI / Client)

- Build profile report and user block interaction buttons.
- Create user feedback submission dialog modal.
- Design in-app support ticketing view for resolution tracking.

Backend Lead (Server / DB)

- Build automated profanity and inappropriate photo content filter.
- Create moderation API endpoint allowing admins to suspend bad actors.
- Implement error report aggregator sending alerts to team webhook.

HIGH-CONCURRENCY DATABASE & CACHING ARCHITECTURE (1,000 USERS)



STAGE 12: ANALYTICS, METRICS & KEY PERFORMANCE INDICATORS

Frontend Lead (UI / Client)

- Integrate privacy-focused product analytics tracking (PostHog).
- Track funnel conversion from profile view to right swipe and match.
- Build client-side performance timing telemetry logging.

Backend Lead (Server / DB)

- Build internal statistics dashboard API query aggregation engine.
- Track core KPIs: Match Conversion Rate, Active Exchanges, Daily Blogs.
- Implement automated daily database backup and health monitoring.

STAGE 13: CONTINUOUS MAINTENANCE & PLATFORM EVOLUTION

Frontend Lead (UI / Client)

- Conduct bi-weekly UI maintenance and dependency version upgrades.
- Iterate on swipe deck UX based on user feedback telemetry.
- Implement dynamic dark theme overrides and new avatar framing options.

Backend Lead (Server / DB)

- Execute monthly security patch updates and API rate limit tuning.
- Optimize database query execution plans based on slow query logs.
- Plan v2 architecture migration for group video skill workshops.

PRODUCT KPI TELEMETRY & SYSTEM HEALTH MONITORING ARCHITECTURE

